



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

PHYSICS

PAPER 3 Practical Test

4023/3

2 hours

SPECIMEN PAPER

INSTRUCTIONS TO SUPERVISORS

Great care should be taken that any confidential information given does not reach the candidates either directly or indirectly.

Prior to the examination, the Supervisor should ensure that all apparatus provided for the candidate as listed in these instructions, are in good working condition.

No access to the Question Paper is allowed before the examination Date and Time.

The Supervisor is NOT allowed to access the Question paper before the examination Date and Time.

These instructions consist of 6 printed pages.

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Instructions for preparing apparatus

In addition to the usual equipment of a Physics laboratory, candidates will require the apparatus specified on page 3 and 4.

Enough apparatus must be provided to ensure that

- (a) no candidate is kept waiting at the end of the allocated time for each question,
- (b) spare apparatus is immediately available in the event of failure or breakage of any item.

Instructions for the Practical Physics Supervisor

- 1 The Supervisor should give details to the examiner if the apparatus used differ(s) in any significant way from that specified in these instructions.**
- 2 Supervisors are asked to provide the information required by the examiner (if any) on page 5 and 6. Pages 5 and 6 should be detached and attached to the 'Supervisor's Results' before despatch to the Examiner.
- 3 The Supervisor must perform the experiments and submit with the candidates' answers, a sample set of results in the same form as that expected of the candidates.**

Question 1

Instructions to supervisor

Apparatus per candidate

2 × G-clamps

1 × 12 cm unstretched slinky spring, diameter $\simeq$ 5 cm

1 × stopwatch (0.01s)

1 × metre rule

1 × electronic balance (can be shared)

Question 2

Instructions to supervisor

Apparatus per candidate:

1 × 250 ml beaker

NB: The supervisor should provide 5 soil samples of same mass with different moisture content e.g.

sample A = 0 % moisture content

sample B = 20 % moisture content

sample C = 40 % moisture content

sample D = 60 % moisture content

sample E = 80 % moisture content

The supervisor to program the microprocessor using a suitable program.

1 × moisture sensor

1 × Arduino UNO microprocessor (PROGRAMMED)

1 × 5 V dc power supply with USB port

1 × 220 Ω resistor

1 × LED

1 × breadboard

10 × breadboard wires (varied)

1 × digital millivoltmeter (0 – 200mV)

5 × connecting wires (mixed)

This form must be completed and returned with the scripts.

REPORT ON THE PHYSICS PRACTICAL EXAMINATION

1. Please submit details of the results obtained by the Supervisor for both questions on a spare copy of question paper clearly marked ‘Supervisor’s Results’ and showing the Centre number.
2. The Supervisor is asked to give the following details using the space provided on page 6 where necessary.
 - (a) Experiment results of the supervisor recorded on a spare copy of the question paper. It is to be clearly labelled “Supervisor’s Results” with the centre number.
 - (b) Any help given to a candidate.
 - (c) Any general difficulties encountered in preparing the apparatus.
 - (d) Any difficulties experienced by particular candidates. This should include reference to difficulties due to faulty apparatus or materials and accidental damage to apparatus or materials. Candidates should be identified by name and index number.
3. Other cases of hardship, such as disability or illness, should be reported to ZIMSEC **in the normal way**.
4. The Supervisor is asked to provide a plan of the work benches, giving details by index numbers of the places occupied by the candidates for each session. The plan should be enclosed with the question papers.
5. **Declaration to be signed by the Supervisor**

The preparation for this practical examination has been carried out so as to fully maintain the security of the examination.

Signed _____

Name (block capitals) _____

Centre number _____

Centre name _____

This form must be completed and returned with the scripts.

Report on any difficulties by candidates

Signature _____

Date _____